

P1072R7

basic_string::resize_and_overwrite

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This version:

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Project:

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Abstract

Optimized writing of data into a basic_string.

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§ 1. Motivation

Performance sensitive code is impacted by the cost of initializing and manipulating strings. When writing data into a `basic_string`, a programmer is faced with an unhappy choice:

- **Pay for extra initialization** — `resize`, which zero initializes, followed by copy.
- **Pay for extra copies** — Populate a temporary buffer, copy it to the string.
- **Pay for extra "bookkeeping"** — `reserve` followed by small appends, each of which checks capacity and null terminates.

The effects of these unhappy choices can all be measured at scale, yet C++'s hallmark is to leave no room between the language (or in this case the library) and another language.

LEWGI polled on this at the [\[SAN\]](#) Meeting:

We should promise more committee time to [\[P1072R1\]](#), knowing that our time is scarce and this will leave less time for other work?

Unanimous consent

LEWG at the [\[SAN\]](#) Meeting:

We want to solve this problem

SF	F	N	A	SA
17	5	0	0	0

Willing to solve this for string without solving it for vector

SF	F	N	A	SA
6	9	2	1	0

§ 2. Proposal

This proposal addresses the problem by adding `basic_string::resize_and_overwrite`:

```
template<typename Operation>
constexpr void resize_and_overwrite(size_type n, Operation op);
```

7. *Effects*: Alters the value of `*this` as follows:

1. — If `n <= size()`, erases the last `size() - n` elements.
2. — If `n > size()`, appends `n - size()` default-initialized elements.

Invokes `erase(begin() + op(data(), n), end())`.

8. *Remarks*:

If `op` throws the behavior is undefined.

Let `o = size()` before the call to `resize_and_overwrite`.

Let `m = op(data(), n)`.

`m <= n` otherwise the behavior is undefined.

If `m > o`, `op` shall replace ([**expr.ass**]) the values stored in the character array [`data() + o`, `data() + m`). Until replaced, the values may be indeterminate [**basic.indet**] [*Note: `*(data() + o)` may not be `CharT()`. - end note*]

`op` may write to `data() + n + 1`. Any value written will be replaced with `CharT()` after `op` returns. [*Note: This facilitates interoperability with functions that write a trailing null. - end note*]

When `op` is called `*this` is in an unspecified state. `op` shall not bind or in any way access `*this`.

`op` shall not allow its first argument to be accessible after it returns.

In order to enable `resize_and_overwrite`, this proposal makes it implementation-defined whether `basic_string` uses `allocator_traits::construct` and `allocator_traits::destroy` to construct and destroy the "char-like objects" that it controls. See [§ 5.2 Allocator Interaction](#) and [§ 8 Wording](#) below for more details.

§ 3. Implementation

`libc++` includes a private implementation of a prior version of this proposal (based on an earlier revision) and uses it to avoid a dynamic allocation in `std::filesystem` [LIBC++].

§ 4. Examples

§ 4.1. Stamping a Pattern

Consider writing a pattern several times into a string:

```
std::string GeneratePattern(const std::string& pattern, size_t count) {
    std::string ret;

    ret.reserve(pattern.size() * count);
    for (size_t i = 0; i < count; i++) {
        // SUB-OPTIMAL:
        // * Writes 'count' nulls
        // * Updates size and checks for potential resize 'count' times
        ret.append(pattern);
    }

    return ret;
}
```

Alternatively, we could adjust the output string's size to its final size, avoiding the bookkeeping in `append` at the cost of extra initialization:

```
std::string GeneratePattern(const std::string& pattern, size_t count) {
    std::string ret;

    const auto step = pattern.size();
    // SUB-OPTIMAL: We memset step*count bytes only to overwrite them.
    ret.resize(step * count);
    for (size_t i = 0; i < count; i++) {
        // GOOD: No bookkeeping
        memcpy(ret.data() + i * step, pattern.data(), step);
    }
}
```

```

    return ret;
}

```

With this proposal:

```

std::string GeneratePattern(const std::string& pattern, size_t count) {
    std::string ret;

    const auto step = pattern.size();
    // GOOD: No initialization
    ret.resize_and_overwrite(step * count, [&](char* buf, size_t n) {
        for (size_t i = 0; i < count; i++) {
            // GOOD: No bookkeeping
            memcpy(buf + i * step, pattern.data(), step);
        }
        return step * count;
    });

    return ret;
}

```

§ 4.2. Interacting with C

Consider wrapping a C API while working in terms of C++'s `basic_string` vocabulary. We *anticipate* over-allocating, as computation of the *length* of the data is done simultaneously with the computation of the *contents*.

```

extern "C" {
    int compress(void* out, size_t* out_size, const void* in, size_t in_size)
    size_t compressBound(size_t bound);
}

std::string CompressWrapper(std::string_view input) {
    std::string compressed;
    // Compute upper limit of compressed input.
    size_t size = compressBound(input.size());

    // SUB-OPTIMAL: Extra initialization
    compressed.resize(size);
}

```

```

    int ret = compress(&*compressed.begin(), &size, input.data(), input.size());
    if (ret != OK) {
        throw ...some error...
    }

    // Set actual size.
    compressed.erase(size);
    return compressed;
}

```

With this proposal:

```

extern "C" {
    int compress(void* out, size_t* out_size, const void* in, size_t in_size);
    size_t compressBound(size_t bound);
}

std::string CompressWrapper(std::string_view input) {
    std::string compressed;
    // Compute upper limit of compressed input.
    size_t size = compressBound(input.size());
    int ret;

    // GOOD: No initialization
    compressed.resize_and_overwrite(size, [&](char* buf, size_t n) {
        ret = compress(buf, &size, input.data(), input.size());
        return *size;
    });
    if (ret != OK) {
        throw ...some error...
    }
    return compressed;
}

```

§ 5. Design Considerations

§ 5.1. Method vs. Alternatives

During the [\[SAN\]](#) Meeting LEWG expressed a preference for implementing this functionality as a new method on `basic_string` (as proposed in [\[P1072R0\]](#)) rather than a standalone "storage buffer" type (one option in [\[P1072R1\]](#)):

Method on string vs storage_buffer type:

Strong Method	Method	Neutral	Type	Strong Type
9	2	3	2	2

During the [\[KOA\]](#) Meeting, LEWG expressed a preference for not weakening `basic_string`'s invariants after `resize_and_overwrite` returns. Under this revision:

- A default-initialized buffer is *only* accessible to `op`.
- Under penalty of undefined behavior, `op` must assign (replace) the values in resulting buffer. The default initialized portion (`[data()+m, data()+n)`) above is erased and inaccessible to the program without further violation of class invariants.
- This proposal follows the "lambda" option discussed in [\[SAN\]](#).

Several other alternatives were discussed at [\[SAN\]](#), [\[KOA\]](#), and on the reflector. Please see [§ 6 Alternatives Considered](#) below for more details.

§ 5.2. Allocator Interaction

Unlocking the desired optimizations requires *some* change to `basic_string`'s interaction with allocators. This proposal does what we think is the simplest possible change: remove the requirement that `basic_string` call `allocator_traits::construct` or `allocator_traits::destroy`.

This restriction should be acceptable because `basic_string` is defined to only hold "non-array trivial standard-layout" types. [\[strings.general\]](#) p1:

This Clause describes components for manipulating sequences of any non-array trivial standard-layout (6.7) type. Such types are called *char-like types*, and objects of *char-like types* are called *char-like objects* or simply *characters*.

Removing calls to `construct` and `destroy` is compatible with `pmr::basic_string` as long as `uses_allocator_v<charT>` is `false`, which should be the case in practice.

Along the way, this proposal clarifies ambiguous language in `[string.require]` and `[container.requirements.general]` by:

- Stating explicitly that `basic_string` is allocator-aware.
- Introducing the term "allocator-aware" earlier in `[container.requirements.general]`.
- Not attempting to mention other "components" in `[container.requirements.general]`. The other allocator-aware containers (just `basic_string` and `regex::match_results`?) can independently state that they are "allocator-aware" in their own clauses.

libstdc++ and msvc allow strings of non-trivial type. That might force those libraries to continue to support `construct` and `destroy` as "extensions". On the other hand, libc++ disallows non-trivial `charTs`, so any such extensions are non-portable. See godbolt.org.

§ 5.3. Undefined Behavior

`resize_and_overwrite` exposes users to UB if they read indeterminate (`[basic.indet]`) values in the string buffer passed to `op`. Despite this foot-gun, `resize_and_overwrite` is appealing because:

- Uninitialized buffers are not uncommon in C++. See for example `new T[20]` or the recently adopted `make_unique_for_overwrite`.
- Dynamic analyzers like Memory Sanitizer [\[MSan\]](#) and Valgrind [\[Valgrind\]](#) can catch uninitialized reads.

The version of this proposal seen by LEWG in [\[Cologne\]](#) throws `std::range_error` when `op` returns a value greater than `n`. The authors now recommend and LEWG confirmed via [\[LEWGNovTelecon\]](#) to use UB. For the `op` to have written greater than `n` bytes, it would have had to overrun the provided buffer (already UB).

The authors further recommend UB if `op` were to throw. `resize_and_overwrite` is motivated by measured performance improvements that are small individually but significant at scale. Providing exception guarantees would incur overhead that is in tension with the original motivation.

§ 5.4. Bikeshedding

What do we want to call this method?

- `resize_and_overwrite` (as proposed). This is consistent with `make_unique_for_overwrite` [P1973R1]. Unlike the `make_unique` version, this concludes with the buffer being overwritten (as far as `op` specifies), rather than requiring further action be taken.
- `resize_default_init` (as proposed). This is consistent with `make_unique_default_init` [P1020R1], adopted in San Diego. This name was removed from the working paper subsequently and renamed `make_unique_for_overwrite`.
- `resize_uninitialized` (as proposed in R0). "Uninitialized" is different from "default-initialized", which is what we want to specify. Also, `uninitialized` is already used elsewhere in the Library to mean "not constructed at all", as in `uninitialized_copy`; so this would be an overloading of the term.

§ 6. Alternatives Considered

§ 6.1. Tag Type

At the [SAN] Meeting, LEWG showed support for a tag argument type:

Approval (vote for as many as you find acceptable)

13	Go back to <code>resize_uninitialized</code>
15	Do tag type (<code>default_initialize</code>) for <code>c'tor</code> / <code>resize()</code>
12	Continue with <code>storage_buffer</code> (as per R2 of this paper)
7	Crack open with a lambda
7	RAII separate type

For example:

```
std::string GeneratePattern(const std::string& pattern, size_t count) {
    const auto step = pattern.size();

    // GOOD: No initialization
    std::string ret(step * count, std::string::default_init);
    for (size_t i = 0; i < count; i++) {
        // GOOD: No bookkeeping
        memcpy(ret.data() + i * step, pattern.data(), step);
    }
}
```

```
    return ret;
}
```

Benefits:

- A constructor that takes a tag type simplifies some use cases, like the example above.
- Matches existing practice in Boost. See [§ 7.7 Boost](#).

Drawbacks:

- Feature creep / complexity — A tag type invites generalizing for all allocator-aware container types. We agree that this is desirable and even has implementation experience in Boost. However default initialization is not enough. There is also "implicit construction" (see [\[P1010R1\]](#), [\[P0593R2\]](#)) and "relocation" (see [\[P1144R0\]](#), [\[P1029R1\]](#)). Neither of these features are yet in the language. It is too early to generalize. Note that the second poll quoted in [§ 1 Motivation](#) shows support for solving this problem for `[basic_]string` but not `vector`.
- In reflector discussion of `make_unique_default_init [Zero]`, there was a preference for avoiding tag types. The standard library has `copy_backward`, not `copy` with a tag, and `count_if`, rather than `count` with a predicate.

Conclusion:

LEWG should explore tags for allocator-aware containers, but that work should not block near-term enablement of efficient `[std::]string` builders.

§ 6.2. Non-Lambda Approach

In [\[P1072R0\]](#) and [\[P1072R3\]](#) of this proposal, the authors considered a method `resize_default_init` / `resize_uninitialized` which left a default-initialized buffer accessible to users of the `basic_string` instance after the method returned. This method was rejected in [\[KOA\]](#), due to the weakened class invariants.

For illustration:

```
extern "C" {
    int compress(void* out, size_t* out_size, const void* in, size_t in_size)
    size_t compressBound(size_t bound);
}
```

```

std::string CompressWrapper(std::string_view input) {
    std::string compressed;
    // Compute upper limit of compressed input.
    size_t size = compressBound(input.size());

    // GOOD: No initialization
    compressed.resize_default_init(size);
    int ret = compress(&*compressed.begin(), &size, input.data(), input.size());
    if (ret != OK) {
        throw ...some error...
    }

    // Suppose size is the value of size before the call to compress and size'
    // is the value of size after the call to compress.
    //
    // If size' < size, then:
    //   std::cout << compressed[size' + 1]
    // ...triggers a read of uninitialized data.

    // Set actual size.
    compressed.erase(size);
    return compressed;
}

```

§ 6.3. Standalone Type: `storage_buffer`

In [\[P1072R1\]](#), we considered `storage_buffer`, a standalone type providing a `prepare` method (similar to the `resize_uninitialized` method proposed here) and `commit` (to promise, under penalty of UB, that the buffer had been initialized).

At the [\[SAN\]](#) Meeting, this approach received support from LEWGI in light of the [\[post-Rapperswil\]](#) email review indicating support for a distinct type. This approach was rejected by the larger LEWG room in San Diego Meeting Diego.

The proposed type would be move-only.

```

std::string GeneratePattern(const std::string& pattern, size_t count) {
    std::storage_buffer<char> tmp;

    const auto step = pattern.size();

```

```

// GOOD: No initialization
tmp.prepare(step * count + 1);
for (size_t i = 0; i < count; i++) {
    // GOOD: No bookkeeping
    memcpy(tmp.data() + i * step, pattern.data(), step);
}

tmp.commit(step * count);
return std::string(std::move(tmp));
}

```

For purposes of the container API, `size()` corresponds to the *committed* portion of the buffer. This leads to more consistency when working with (and explicitly copying to) other containers via iterators, for example:

```

std::storage_buffer<char> buf;
buf.prepare(100);
*fill in data*
buf.commit(50);

std::string a(buf.begin(), buf.end());
std::string b(std::move(buf));

assert(a == b);

```

Benefits:

- By being move-only, we would not have the risk of copying types with trap representations (thereby triggering UB).
- Uninitialized data is only accessible from the `storage_buffer` type. For an API working with `basic_string`, no invariants are weakened. Crossing an API boundary with a `storage_buffer` is much more obvious than a "`basic_string` with possibly uninitialized data." Uninitialized bytes (the promise made by `commit`) never escape into the valid range of `basic_string`.

Drawbacks:

- `storage_buffer` requires introducing an extra type to the standard library, even though its novel functionality (from `string` and `vector`) is limited to the initialization abilities.

- `basic_string` is often implemented with a short-string optimization (SSO) and an extra type would need to implement that (likely by additional checks when moving to/from the `storage_buffer`) that are often unneeded.

§ 6.4. Externally-Allocated Buffer Injection

In [P1072R1], we considered that `basic_string` could "adopt" an externally `allocator::allocate`'d buffer. At the [SAN] Meeting, we concluded that this was:

- **Not critical**
- **Not constrained in the future**
- **Overly constraining to implementers.** Allowing users to provide their own buffers runs into the "offset problem". Consider an implementation that stores its `size` and `capacity` inline with its data, so `sizeof(container) == sizeof(void*)`.

```
class container {  
    struct Rep {  
        size_t size;  
        size_t capacity;  
    };  
  
    Rep* rep_;  
};
```

If using a `Rep`-style implementation, the mismatch in offsets requires an $O(N)$ move to shift the contents into place and trigger a possible reallocation.

- **Introducing new pitfalls.** It would be easy to mix `new[]` and `allocator::allocate` inadvertently.

§ 7. Related Work

§ 7.1. Google

Google has a local extension to `basic_string` called `resize_uninitialized` which is wrapped as `STLStringResizeUninitialized`.

- [Abseil] uses this to avoid bookkeeping overheads in `StrAppend` and `StrCat`.

- [\[Snappy\]](#)
 - In [decompression](#), the final size of the output buffer is known before the contents are ready.
 - During [compression](#), an upperbound on the final compressed size is known, allowing data to be efficiently added to the output buffer (eliding [append](#)'s checks) and the string to be shrunk to its final, correct size.
- [\[Protobuf\]](#) avoids extraneous copies or initialization when the size is known before data is available (especially during parsing or serialization).

§ 7.2. MongoDB

MongoDB has a string builder that could have been implemented in terms of [basic_string](#) as a return value. However, as explained by Mathias Stearn, zero initialization was measured and was too costly. Instead a custom string builder type is used:

E.g.: <https://github.com/mongodb/mongo/blob/master/src/mongo/db/fts/unicode/string.h>

```
/**
 * Strips diacritics and case-folds the utf8 input string, as needed to sup
 * options.
 *
 * The options field specifies what operations to *skip*, so kCaseSensitive
 * means to skip case folding and kDiacriticSensitive means to skip diacrit
 * striping. If both flags are specified, the input utf8 StringData is retu
 * directly without any processing or copying.
 *
 * If processing is performed, the returned StringData will be placed in
 * buffer. buffer's contents (if any) will be replaced. Since we may retur
 * the input unmodified the returned StringData's lifetime is the shorter o
 * the input utf8 and the next modification to buffer. The input utf8 must
 * point into buffer.
 */
static StringData caseFoldAndStripDiacritics(StackBufBuilder* buffer,
                                             StringData utf8,
                                             SubstrMatchOptions options,
                                             CaseFoldMode mode);
```

(Comments re-wrapped.)

§ 7.3. VMware

VMware has an internal string builder implementation that avoids `std::string` due, in part, to `reserve`'s zero-writing behavior. This is similar in spirit to the MongoDB example above.

§ 7.4. Discussion on std-proposals

This topic was discussed in 2013 on std-proposals in a thread titled "Add `basic_string::resize_uninitialized` (or a similar mechanism)":

<https://groups.google.com/a/isocpp.org/forum/#!topic/std-proposals/XIO4KbBTxl0>

§ 7.5. DynamicBuffer

The [\[N4734\]](#) (the Networking TS) has *dynamic buffer* types.

§ 7.6. P1020R1

See also [\[P1020R1\]](#) "Smart pointer creation functions for default initialization". Adopted in San Diego.

§ 7.7. Boost

Boost provides a related optimization for vector-like containers, introduced in [\[SVN r85964\]](#) by Ion Gaztañaga.

E.g.: [boost/container/vector.hpp](#):

```
//! <b>Effects</b>: Constructs a vector that will use a copy of allocator a
//!    and inserts n default initialized values.
//!
//! <b>Throws</b>: If allocator_type's allocation
//!    throws or T's default initialization throws.
//!
//! <b>Complexity</b>: Linear to n.
//!
//! <b>Note</b>: Non-standard extension
vector(size_type n, default_init_t);
```



```
vector(size_type n, default_init_t, const allocator_type &a)
...
void resize(size_type new_size, default_init_t);
...
```

These optimizations are also supported in Boost Container's `small_vector`, `static_vector`, `deque`, `stable_vector`, and `string`.

§ 7.8. Thrust

The Thrust library has "a RAII-type `thrust::detail::temporary_array` which has a vector-like interface and a constructor with a tag parameter that indicates its elements should not be initialized." - [Bryce Adelstein Leibach].

E.g. `thrust/thrust/detail/temporary_array.inl`:

```
template<typename T, typename TemporaryArray, typename Size>
__host__ __device__
typename thrust::detail::disable_if<
    avoid_initialization<T>::value
>::type
    construct_values(TemporaryArray &a,
                    Size n)
{
    a.default_construct_n(a.begin(), n);
} // end construct_values()
```

§ 8. Wording

Wording is relative to [\[N4878\]](#).

Motivation for some of these edits can be found in [§ 5.2 Allocator Interaction](#).

§ 8.1. [basic.string]

In [basic.string], in the synopsis, add `resize_for_overwrite`:

```
...
// 21.3.3.5, capacity
constexpr size_type size() const noexcept;
constexpr size_type length() const noexcept;
constexpr size_type max_size() const noexcept;
constexpr void resize(size_type n, charT c);
constexpr void resize(size_type n);
template<typename Operation>
constexpr void resize_and_overwrite(size_type n, Operation op);
constexpr size_type capacity() const noexcept;
constexpr void reserve(size_type res_arg);
constexpr void shrink_to_fit();
constexpr void clear() noexcept;
[[nodiscard]] constexpr bool empty() const noexcept;
```

In [string.require] clarify that `basic_string` is an allocator-aware container, and then add an exception for `construct` and `destroy`.

3. In every specialization `basic_string<charT, traits, Allocator>`, the type `allocator_traits<Allocator>::value_type` shall name the same type as `charT`.
~~Every object of type `basic_string<charT, traits, Allocator>` uses an object of type `Allocator` to allocate and free storage for the contained `charT` objects as needed. The `Allocator` object used is obtained as described in [container.requirements.general]. In every specialization `basic_string<charT, traits, Allocator>`, and the type `traits` shall satisfy the character traits requirements ([char.traits]). [Note: The program is ill-formed if `traits::char_type` is not the same type as `charT`. — end note]~~
4. `basic_string` is an allocator-aware container as described in [container.requirements.general], except that it is implementation-defined whether `allocator_traits::construct` and `allocator_traits::destroy` are used to construct and destroy the contained `charT` objects.
5. References, pointers, and iterators referring to the elements of a `basic_string` sequence may be invalidated by the following uses of that `basic_string` object:
...

In `[string.capacity]`:

```
constexpr void resize(size_type n, charT c);
```

5. *Effects*: Alters the value of `*this` as follows:

1. — If `n <= size()`, erases the last `size() - n` elements.
2. — If `n > size()`, appends `n - size()` copies of `c`.

```
constexpr void resize(size_type n);
```

6. *Effects*: Equivalent to `resize(n, charT())`.

```
template<typename Operation>
```

```
constexpr void resize_and_overwrite(size_type n, Operation op);
```

7. *Effects*: Alters the value of `*this` as follows:

1. — If `n <= size()`, erases the last `size() - n` elements.
2. — If `n > size()`, appends `n - size()` default-initialized elements.

Invokes `erase(begin() + op(data(), n), end())`.

8. *Remarks*:

If `op` throws the behavior is undefined.

Let `o = size()` before the call to `resize_and_overwrite`.

Let `m = op(data(), n)`.

`m <= n` otherwise the behavior is undefined.

If `m > o`, `op` shall replace (`[expr.ass]`) the values stored in the character array `[data() + o, data() + m)`. Until replaced, the values may be indeterminate [**basic.indet**]. [*Note: `*(data() + o)` may not be `CharT()`. - end note*]

`op` may write to `data() + n + 1`. Any value written will be replaced with `CharT()` after `op` returns. [*Note: This facilitates interoperability with functions that write a trailing null. - end note*]

When `op` is called `*this` is in an unspecified state. `op` shall not bind or in any way access `*this`.

op shall not allow its first argument to be accessible after it returns.

```
constexpr size_type capacity() const noexcept;  
...
```

§ 8.2. [container.requirements.general]

In [container.requirements.general] clarify the ambiguous "components affected by this subclause" terminology in p3. Just say "allocator-aware containers".

3. All of the containers defined in this Clause except array are allocator-aware containers. ~~For the components affected by this subclause that declare an allocator_type~~ Objects ~~objects~~ stored in ~~these components~~ allocator-aware containers, unless otherwise specified, shall be constructed using the function
allocator_traits<allocator_type>::rebind_traits<U>::construct and
destroyed using the function
allocator_traits<allocator_type>::rebind_traits<U>::destroy (20.10.9.3),
where U is either allocator_type::value_type or an internal type used by the container.
...

We can then simplify p15:

15. ~~All of the containers defined in this Clause and in 21.3.3 except array meet the additional requirements of an allocator-aware container, as~~ Allocator-aware containers meet the additional requirements described in Table 78.

§ 9. History

§ 9.1. R6 → R7

- Rebased on [N4878].
- Fixed CompressWrapper example.
- Allowed writing one past the end.

§ 9.2. R5 → R6

LEWG reviewed R5 via [\[LEWGNovTelecon\]](#).

POLL: `resize_default_init` should be able to work in `constexpr` context.

No objection to unanimous consent. Attendance: 23

POLL: We prefer UB over throwing an exception if the operation reports it wrote more bytes than it has access to.

No objection to unanimous consent. Attendance: 21

POLL: We want to rename `resize_default_init` for consistency, in a mailing list review.

SF	F	N	A	SA
3	12	0	1	0

Modifications since R5:

- Indicated poll outcomes for UB when `op` reports an invalid value.
- Renamed to `resize_and_overwrite`. This is based on naming discussions for `std::make_unique_default_init` being ultimately renamed to `std::make_unique_for_overwrite`, but differs slightly because the memory accessible has been overwritten. Overwriting the buffer is an integral part of the operation performed.
- Applied `constexpr`

§ 9.3. R4 → R5

A draft of this revision was presented to LEWG at the [\[Cologne\]](#) meeting.

Forward to LWG for C++23

SF	F	N	A	SA
10	5	1	0	0

- Rebased on [\[N4830\]](#).
- Proposed UB instead of `throwing std::range_error`.

§ 9.4. R3 → R4

- Applied feedback from the [\[KOA\]](#) meeting.
- Moved to using a callback routine, rather than leaving invariants weakened from the time `resize_default_init` returned until values in the buffer were assigned.

§ 9.5. R2 → R3

- Applied Jonathan Wakely's editorial comments on [§ 8 Wording](#).
- Rebased on [\[N4791\]](#).
- Editorial changes to [§ 1 Motivation](#) and [§ 6.1 Tag Type](#) to support our design choice.
- Added the reference to [\[LIBC++\]](#) in [§ 3 Implementation](#).

§ 9.6. R1 → R2

Applied feedback from [\[SAN\]](#) Meeting reviews.

- Reverted design to "option A" proposed in [\[P1072R0\]](#).
- Switched from `resize_uninitialized` to `resize_default_init`.
- Added discussion of alternatives considered.
- Specified allocator interaction.
- Added wording.

§ 9.7. R0 → R1

Applied feedback from LEWG [\[post-Rapperswil\]](#) Email Review:

- Shifted to a new vocabulary types: `storage_buffer` / `storage_node`
 - Added presentation of `storage_buffer` as a new container type
 - Added presentation of `storage_node` as a node-like type
- Added discussion of design and implementation considerations.
- Most of [\[P1010R1\]](#) Was merged into this paper.

§ 10. Acknowledgments

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